

Decision-dependent distributionally robust optimization for resilient and sustainable multimodal transportation under endogenous uncertainty

1 Order data

Table 1: Origins, destinations and demands of different orders within the transport network

Order No.	$o(l)$	$d(l)$	Required demand (tonne)	g_l (TEU container)	y_l (hour)
1	5	3	212	15	70
2	10	9	1182	84	80
3	2	10	794	57	150
4	5	4	128	9	150
5	2	9	182	13	150
6	4	8	99	7	110
7	6	3	150	11	70
8	6	1	343	25	200
9	4	9	168	12	70
10	3	7	567	41	230
11	9	11	960	69	40
12	1	3	240	17	230
13	2	4	790	56	110
14	8	6	83	6	110
15	8	7	570	41	190
16	8	11	150	11	160
17	4	2	1220	87	110
18	11	7	110	8	40
19	7	8	350	25	190
20	9	8	410	29	200
21	3	2	300	21	260
22	5	2	130	9	260
23	10	8	225	16	120
24	5	6	850	61	130
25	8	6	500	36	110
26	2	1	275	20	40
27	7	4	150	11	110
28	11	10	800	57	40
29	5	3	175	13	70
30	11	9	87	6	40

2 Parameters data

Table 2: Input data for D³RTM

Notation	data		Notation	data		Notation	road	rail	waterway
$\bar{c}_j$	194.74		ϕ_{no}	0.36		f_{ij}^m	1000	1000	3000
Γ	5000		ϕ_{co}	0.17		$\tilde{c}_{ij}^m$	50	50	150
τ_j	4		ϕ_{ac}	0.09		b^m	2	28	212
$\bar{e}_j$	11.27		ϕ_{rn}	0.25		e^m	1.4	0.5	0.36
ϕ_{em}	0.05		ϕ_{wn}	0.06		$\hat{e}^m$	10	5	3